AC Gray Box

Y22 (2022) — English translation

There is no need to estimate errors in this problem! This problem corresponds to the program PE3.

This problem asks you to examine a gray box with four terminals. Inside the box there is a known circuit (see figure). A sinusoidal voltage of amplitude 1 V and adjustable frequency ω is applied to the outputs AB. You can measure the amplitudes of the total current I through the circuit (between outputs AB) and the voltage U across the section CD. In addition, you can set the resistance value of the variable resistor R_1. The program displays graphs of I (in amperes) and U (in volts) over the selected frequency range.

The program takes as input the frequency range ω (in s^{-1}) and the resistance R_1 (in Ω). To set the range, you need to specify the minimum frequency, the maximum frequency, and the step used to plot the graph. All frequencies must be positive, and the maximum must be greater than the minimum! The integer part is separated from the fractional part by a decimal point.

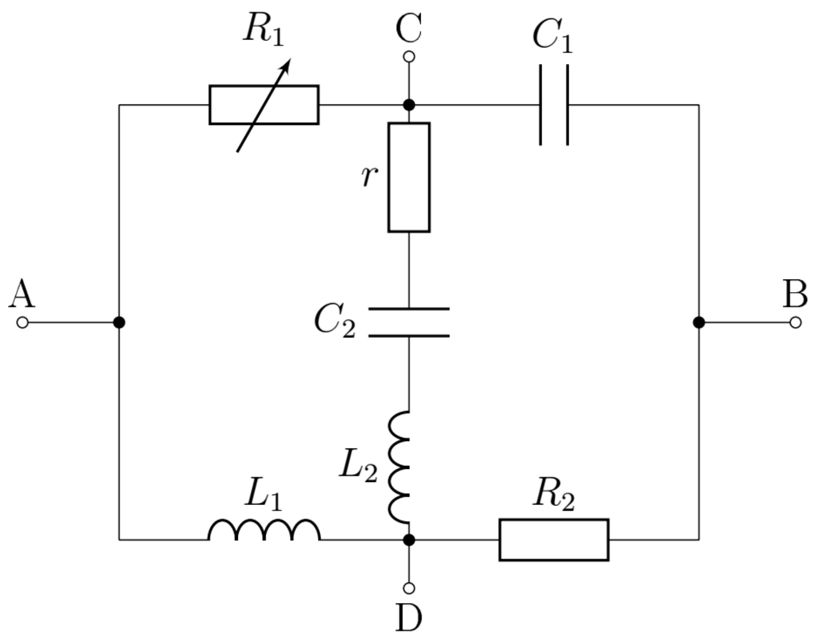

Figure 1: Figure 1.

A1	Define the circuit parameters (L_1, L_2, C_1, C_2, R_2, r).	**7.00**

Source: `https://pho.rs/p/2849`